

any forbidden stem or lemma. If a match is found, the method backtracks to the first token of the offending word and re-samples, blocking only that specific token at that position. For *accipicchia*, the online approach evaluates each generated word against the forbidden forms at runtime, and only intervenes if a word such as *meraviglia* or *caspita* is actually produced — leaving *per*, *la*, and all other innocent tokens freely available. Across the same 194 items, the model produces on average 19.3 words per output, all of which are checked at runtime — yet zero violations are recorded, confirming that the online constraint intervenes surgically only when needed rather than preemptively suppressing large portions of the vocabulary.

C. Appendix C: SAE implementation details

SAE configuration We use JumpReLU SAE from `google/gemma-scope-2-4b-it` [20], applied to the residual stream of `google/gemma-3-4b-it`, which has 34 transformers layers in its text module. The Gemma Scope 2 checkpoints provides SAEs at four depths: layer 9, 17, 22 and 29 (approximately 25%, 50%, 65% and 85% of network depth). We select layer 29, as later layers are known to carry richer semantic representation compared to early layers [28], making them better suited for identifying concept-level features associated with specific words. The SAE uses a dictionary width of 65k features and medium L0 sparsity, with weights loaded in `bfloat16` and kept frozen throughout. Feature identification runs once per forbidden word per card and is cached to avoid redundant forward passes across items sharing the same forbidden word. The feature prompt template "*La parola vietata è: {word}*" is tokenized with special tokens; the token span of w_i is located by matching the tokenized word (with and without a leading space) against the full input sequence, falling back to the final token if no match is found.

D. Appendix E: Failure Patterns in Model-Generated Descriptions

Beyond aggregate accuracy, qualitative inspection of the guessing phase reveals several recurring failure patterns in model-generated descriptions.

Code-switching While descriptions generated under the prompting condition remain consistently in Italian, constrained generation introduces a small but noticeable rate of language shifts (6.35%), which becomes far more pronounced under the SAE condition (44.83%), with English, French, and Spanish being the most frequent alternates. This suggests that intervening on internal representations disrupts the model’s linguistic behaviour in ways that explicit lexical guidance does not.

Referential failure A significant proportion of descriptions are nonsensical or semantically detached from the target word, making guessing effectively impossible. Out of 160 ground-truth entries evaluated, 25 (15.62%) were flagged as containing an inappropriate description. A striking example is the word *semolino* (semolina), described under two conditions by the same model as follows:

- `gemma_sae`: *“Piccola protuberanza dura e rotonda, spesso presente sulla pelle, solitamente causata da irritazione o crescita di foruncoli.”* (“A small, hard, round bump, often found on the skin, usually caused by irritation or the growth of pimples.”)
- `gemma_constrained`: *“Piccola goccia di liquido, spesso salata, che si stacca da una fonte, come una lacrima o una pioggia leggera.”* (“A small drop of liquid, often salty, that detaches from a source, such as a tear or light rain.”)

Neither description bears any semantic relation to the target word: the generated text is fluent and well-formed, but lacks any associative link to the intended referent.

Category mismatch In several cases, the model describes a semantically related but categorically distinct concept — for instance, foregrounding the activity rather than the agent, or the domain rather than the practitioner. An example is the word *falegname* (carpenter), described by `gpt_think_constrained` as:

- `gpt_think_constrained`: *“Arte che trasforma il legno: taglia, intaglia, costruisce mobili e strutture con seghe, scalpelli e martelli.”* (“The art of transforming wood: cutting, carving, building furniture and structures with saws, chisels and hammers.”)

By framing the description around the craft (*arte*) rather than the practitioner, the model effectively steers the guesser away from the intended referent.

E. Tables

Note on evaluation metrics. The metrics reported in the Eval section of each table are independent counters and do not sum to 100%. A single output may simultaneously trigger multiple error categories (e.g., a target leak and a filter failure), so the categories are not mutually exclusive.

Section	Metric	Prompting	Constrained		Manip.	Baseline
			a priori	online	SAEs	no forb.
Eval	True accuracy	91.20	98.50	96.40	55.70	34.00
	% exact viol.	2.60	0.00	0.00	31.40	52.10
	% only morphol. viol.	4.60	0.00	0.00	11.90	12.30
	% no <eot>	0.00	0.00	0.00	0.00	0.00
	% target leaked	1.50	1.50	3.60	1.00	1.50
gpt-oss judge	pass@1	0.00	0.00	0.00	0.00	0.00
	pass@2	0.00	0.50	0.50	0.00	0.50
	pass@3	1.00	1.50	2.60	1.00	3.60
	pass@5	1.50	3.60	5.20	3.10	5.70
	pass@10	3.60	7.70	8.80	5.70	9.30
	min	4e-6	4e-6	4e-6	1e-6	4e-6
	p25	1.1e-5	3.0e-5	2.6e-5	1.0e-5	2.7e-5
	median	1.9e-5	6.9e-5	6.2e-5	1.9e-5	1.2e-4
	avg	1.9e-3	3.4e-3	4.5e-3	1.8e-3	5.0e-3
	p75	1.3e-4	3.5e-4	6.1e-4	1.6e-4	1.3e-3
	max	0.1143	0.1564	0.1765	0.0576	0.1437
gemma judge	pass@1	0.00	0.00	0.00	0.00	0.00
	pass@2	1.50	2.10	4.60	4.10	3.10
	pass@3	4.10	4.60	7.70	7.20	7.20
	pass@5	6.20	9.80	9.80	13.90	11.90
	pass@10	11.30	17.50	17.00	21.10	24.70
	min	0	0	0	0	0
	p25	0	0	0	0	0
	median	0	0	0	0	0
	avg	6.9e-4	0	1.4e-4	4.5e-4	1.9e-4
	p75	4e-6	3e-6	2e-6	1.2e-5	2e-6
	max	0.1272	3.2e-3	1.8e-2	4.7e-2	1.8e-2

Table 4
Evaluation results – **gemma-3-4b-it**

Section	Metric	Prompting	Constrained		Baseline
			a priori	online	no forb.
Eval	True accuracy	72.20	82.50	82.00	24.70
	% exact viol.	16.00	0.00	0.00	55.20
	% only morphol. viol.	8.70	0.00	0.50	14.90
	% no <eot>	0.00	0.00	0.00	0.00
	% target leaked	3.10	17.50	17.50	12.90
gpt-oss judge	pass@1	0.00	0.00	0.00	0.00
	pass@2	0.00	0.00	0.00	0.00
	pass@3	0.50	0.50	0.50	1.00
	pass@5	1.00	0.50	0.50	2.60
	pass@10	3.60	2.10	2.10	6.20
	min	2e-6	2e-6	1e-6	1e-6
	p25	1.1e-5	1.1e-5	1.1e-5	1.2e-5
	median	2.5e-5	1.6e-5	1.5e-5	3.2e-5
	avg	8.6e-4	8.7e-5	9.0e-5	1.8e-3
	p75	1.4e-4	3.0e-5	2.6e-5	3.9e-4
	max	0.0635	3.3e-3	5.6e-3	0.1416
gemma judge	pass@1	0.00	0.00	0.00	0.00
	pass@2	2.10	6.70	6.70	3.10
	pass@3	4.10	11.30	13.90	7.70
	pass@5	10.30	17.50	19.10	11.30
	pass@10	18.60	24.20	24.70	17.50
	min	0	0	0	0
	p25	0	0	0	0
	median	0	0	0	0
	avg	4.4e-4	2.8e-3	1.1e-3	1.1e-3
	p75	4e-6	3.7e-5	4.3e-5	4e-6
	max	0.0366	0.2456	0.0498	0.1358

Table 5
Evaluation results – **gpt-oss-20b (no reasoning)**